



CMPE-211

OBJECT ORIENTED PROGRAMMING

LAB MANUAL

Release V20

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CMPE – 211

Preliminary Work (Pre-Lab Activity)

Laboratory Experiment #1

Textbook Material: [Chapter 1](#) «Introduction to Java»
 [Chapter 2](#) «Elementary Programming»

• • • Tutorial

The Tutorial covers introduction to **Eclipse** IDE, which is used in all following laboratory experiments. Tutorial is conducted by one of the Course Lab. assistants who guides Your laboratory group. Explanations are based on the **Supplement 2f**.

Main topics to cover during 1-hour tutorial:

- how to Eclipse **I**ntegrated **D**evelopment **E**nvironment (**IDE**),
- how to create, compile and execute Java **console** application,
- how to use **debugger** to locate logical errors in programs,

All these skills are essential for performing successfully upcoming lab tasks.

Read the supplement in advance (before entering the Lab).

• • • Example

☀ Write (copy), Modify, Compile, Execute and Debug **Java program** covered by the following example:

Example

```
import java.util.Scanner;

public class RepeatAdditionQuiz {
    public static void main(String[] args) {
        int number1 = (int)(Math.random() * 10);
        int number2 = (int)(Math.random() * 10);

        // Create a Scanner
        Scanner input = new Scanner(System.in);

        System.out.print(
            "What is " + number1 + " + " + number2 + "? ");
        int answer = input.nextInt();

        while (number1 + number2 != answer) {
            System.out.print("Wrong answer. Try again. What is "
                + number1 + " + " + number2 + "? ");
            answer = input.nextInt();
        }

        System.out.println("You got it!");
    }
}
```

• • • TASK 1

The constant π that is also known as Archimedes' constant was used by Egyptians but an accurate approximation was not available (it was approximated as 3.1605). In the 5th century AD, Chinese

mathematics approximated it to seven digits, while Indian mathematics made a five-digit approximation, both using geometrical techniques. Around 250BC, the Greek mathematician Archimedes found that it is in the interval (223/71, 22/7). Archimedes knew, what so many people to this day do not, that it is not equal 22/7, and made no claim to have discovered the exact value. The 14 correct digits were computed in 15th century and 527 correct digits were computed in 19th century. Write, Compile and Execute a Java program that approximately calculates π using the following expression, by considering 15 terms:

$$\pi = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + \frac{4}{13} + \dots$$

• • • TASK 2

Write, Compile and Execute a Java program that reads an investment amount, annual interest rate and number of years and displays the future investment value using the following formula:

$$\text{futureInvestmentValue} = \text{investmentAmount} \times (1 + \text{monthlyInterestRate})^{\text{numberOfYears} \times 12}$$

For example, if you enter amount 1000, annual interest rate 3.25% and the number of years 1, the future value is 1032.98. Here is a sample run:

```
Enter investment amount: 1000.56 ↵ Enter
Enter annual interest rate in percentage: 4.25 ↵ Enter
Enter number of years: 1 ↵ Enter
Accumulated value is $1043.92
```

- **NOTE:** Use `pow` method in `Math` class (see page 70).

• • • Sources

- Introduction to Java Programming and Data Structures, Comprehensive Version, 11th Edition, Y. D. Liang, Pearson, 2019.